

Joseph-Vu Trinh

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Education

Virginia Tech

Bachelor of Science - Computer Science

Blacksburg, VA

Graduating May 2028

Relevant Coursework: Data Structures & Algorithms, Computer Systems, Computer Organization, Object-Oriented Programming

Experience

Software Developer

June 2026 – Present

Virginia Tech Department of Computer Science

Blacksburg, VA

- Developing backend REST API services in **Python** and **Flask** for DBWorkout, a SQL practice platform deployed to **170** students in Virginia Tech's CS4604 database course
- Implementing server-side SQL execution and validation logic against isolated **PostgreSQL** sandbox environments via **SQLAlchemy**, executing student-submitted queries and comparing result sets against expected output
- Integrating the **OpenAI API** to generate contextual hints from a student's failed query and the exercise schema, part of a feedback overhaul after only **42%** of students found initial feedback actionable

Full Stack Developer Intern

Sep 2024 – Dec 2024

Engauge

College Park, MD

- Shipped a real-time collaborative whiteboard supporting live communication across **30+** concurrent clients using **Firestore** listeners, contributing to a **30%** increase in student engagement during live classroom sessions
- Engineered a **Firestore** data model that scoped whiteboards to individual students while providing instructors real-time visibility across session submissions
- Rebuilt the front-end component architecture of a classroom web application in **React** and **TypeScript**, optimizing render performance for **150+** students during live sessions

Undergraduate Researcher

Jan 2026 – June 2026

Virginia Tech Department of Computer Science

Blacksburg, VA

- Engineered components of a machine learning pipeline using **OpenPose** and gaze estimation to classify student engagement from classroom videos, improving classification accuracy by **25%** across **150+** students
- Built the feature extraction stage of the pipeline, processing 180 one-minute video segments to extract and track pose and gaze features across recordings
- Validated model outputs against survey-based observations, debugging inconsistencies to improve pipeline reliability

Web Developer

July 2026 – Present

SASE at Virginia Tech

Blacksburg, VA

- Refactored the organization's **React** codebase by migrating legacy CSS to **Tailwind CSS**, establishing a responsive, mobile-first front-end architecture and improving maintainability
- Collaborated with organization leadership to gather requirements, iterate on designs, and deliver a production-ready web application ahead of deployment

Projects

Professor Search Platform | VTHacks | Python, React, Databricks Agent Bricks, GraphQL

- Engineered Python data pipelines with asynchronous Reddit scraping and paginated **GraphQL** ingestion, applying filtering, deduplication, retry logic, and rate-limit handling to sustain throughput across **16,000+** records spanning **577 professors** and **3,267 reviews**
- Integrated the curated dataset with **Databricks Agent Bricks** for an LLM-powered recommendation system answering natural-language questions about professors and courses, and developed the supporting React interface

Playlist Curator | React, TypeScript, FastAPI, Anthropic API, Spotify Web API, librosa

- Developed an AI-powered playlist curation platform using **React**, **TypeScript**, **FastAPI**, and the **Spotify Web API** that evaluates tracks against user-defined themes before presenting a GitHub-style diff for users to review and apply changes to live playlists
- Integrated **10+ backend API endpoints** and local audio signal analysis with librosa to analyze playlists of arbitrary size using both acoustic features and Spotify metadata

Technical Skills

Languages: Python, Java, C/C++, C#, SQL, JavaScript, TypeScript, Swift

Frameworks: Flask, FastAPI, React, Node.js, Tailwind CSS, React Native, SwiftUI

Tools & Technologies: Git, Linux/Unix, Figma, PostgreSQL, Docker, CI/CD, REST APIs, AWS, Firebase/Firestore, MongoDB (NoSQL)